

## **CSCI 112 Project – Tic Tac Toe Game**

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## **Introduction:**

Tic Tac Toe is a very popular paper-pencil game often played in classrooms on the last page of the notebook. In this game, two players mark X or O one by one in a 3x3 grid. To win the game, one has to complete a pair of 3 symbols in a line, and that can be a horizontal line, a vertical line, or a diagonal line.

The objective of this program is to create a tic tac toe game. The game is divided into two functions: Human vs Human and Human vs Computer. The players choose the mode of game they wish to play. The player who first wins 3 trials wins the game. The player enters the row and column number of the position they want to insert their symbol in.

## **Code:**

The following is the complete code of our program:

```
#include <iostream>
#include <iomanip>
#include <cstdlib> //standard header file that defines a list of functions
using namespace std;
int ComputerRow, ComputerCol; // The row and column number chosen by the computer
int PlayerRow, PlayerCol; // The row and column number chosen by the player

char board[3][3] = { { '?', '?', '?' }, { '?', '?', '?' }, { '?', '?', '?' } }; //2d-array to
show the base of the game
int scoreofX = 0; // Counts the cumulative score of Player X through all trials
(HumanvsComputer function)
int scoreofO = 0; // Counts the cumulative score of Player O through all trials
(HumanvsComputer function)
bool player[3][3], computer[3][3]; // additional arrays for Human vs Computer function
int row = 0, column = 0, choice;
```

```

char mark = 'X';
bool tie; // returned value determines if game is tied or not

void Display_Board()
{
    cout << " 1 2 3" << endl; //we put 1,2,3 to show this is the header column

    for (int i = 1; i <= 3; i++) //loop for row header
    {
        cout << i << " "; //number of rows printed

        for (int j = 1; j <= 3; j++) //loop for the base of the game
        {
            cout << board[i - 1][j - 1] << " "; //subtract -1 to convert it to
array values
        }

        cout << endl; //output a new line
    }
}

void HumanvsHuman_check() //players turn
{
    int user_row, user_column;

    if (mark == 'X') //if player X turn then we cout this message for them to choose
row and column
    {
        cout << "Choose where to insert your 'X' mark, insert row then column: ";

    }

    else if (mark == 'O') //if player O turn then we cout this message for them to
choose row and column
    {
        cout << "Choose where to insert your 'O' mark, insert row then column: ";

    }

    cin >> user_row >> user_column;

    while (user_row < 1 || user_row > 3 || user_column < 1 || user_column > 3)
    { //to check if the user have entered valid row and column number if not this
message will be shown

        cout << "Choose a valid row and column between 1 and 3, try again: ";
        cin >> user_row >> user_column; //user enter again

    }

    row = user_row - 1; //to convert the user row number to array indicies
    column = user_column - 1; //to convert the user column number to array indicies

```

```

        if (board[row][column] != '?')//to check if the user selected the same position as
other player
        {
            cout << "Already selected, try another position" << endl;
            return;//it goes back to ask the user again to enter row and column number
        }

        if (mark == 'X') //when it is the player X turn the program will print X at
chosen position
        {
            board[row][column] = 'X';
            mark = 'O';//the next turn will be given to player O
        }
        else if (mark == 'O') //when the player O turn the program will print O at chosen
position
        {
            board[row][column] = 'O';
            mark = 'X';//the next turn will be given to player X
        }

    }

bool HumanvsHuman_check_win()//to check if the players won or not
{
    for (int i = 0; i < 3; i++) //each horizontal and vertical row and column
    {

        if (board[i][0] == board[i][1] && board[i][1] == board[i][2] &&
board[i][0] != '?')
        {
            return true;//to check horizontal win
        }

        if (board[0][i] == board[1][i] && board[1][i] == board[2][i] &&
board[0][i] != '?')
        {
            return true;//to check vertical win
        }

        if ((board[0][0] == board[1][1] && board[1][1] == board[2][2] && board[0][0] !=
'?') || (board[0][2] == board[1][1] && board[1][1] == board[2][0] && board[0][2] != '?'))
        {
            return true;//to check diagonal win
        }

        return false;//no win if none of the statements are met
    }

int check_tie()//to check if there is a tie
{
    for (int i = 0; i < 3; i++)//to check every row
    {
        for (int j = 0; j < 3; j++)//to check every column
        {

```

```

        if (board[i][j] == '?')//if there is a question mark left on the
board
        {
            return 0;//there is no tie
        }
    }
    return 1;//else there is a tie
}

void board_reset()//to reset the board after every new game again
{
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            board[i][j] = '?';
        }
    }
}

int winforX() {
    for (int i = 0; i < 3; i++) //each horizontal and vertical row and column
    {
        if (board[i][0] == board[i][1] && board[i][1] == board[i][2] && board[i][0]
== 'X')
        {
            return 1;//to check horizontal win
        }

        if (board[0][i] == board[1][i] && board[1][i] == board[2][i] && board[0][i]
== 'X')
        {
            return 1;//to check vertical win
        }
    }
    if ((board[0][0] == board[1][1] && board[1][1] == board[2][2] && board[0][0] ==
'X') || (board[0][2] == board[1][1] && board[1][1] == board[2][0] && board[0][2] == 'X'))
    {
        return 1;//to check diagonal win
    }
    return 0;//no win if none of the statements are met
}

int winforO() {
    for (int i = 0; i < 3; i++) //each horizontal and vertical row and column
    {
        if (board[i][0] == board[i][1] && board[i][1] == board[i][2] && board[i][0]
== 'O')
        {
            return 1;//to check horizontal win
        }

        if (board[0][i] == board[1][i] && board[1][i] == board[2][i] && board[0][i]
== 'O')
        {

```

```

        return 1; //to check vertical win
    }
}
if ((board[0][0] == board[1][1] && board[1][1] == board[2][2] && board[0][0] ==
'O') || (board[0][2] == board[1][1] && board[1][1] == board[2][0] && board[0][2] == 'O'))
{
    return 1; //to check diagonal win
}
return 0; //no win if none of the statements are met
}

void printWinOrDraw() { // Based on the return values of previous functions , it outputs
the result of the game.
    if (winforX() == 1) { // If player X wins the game
        cout << "player X won!" << endl;
        scoreofX++; // Increments the score of X
    }
    else if (winforO() == 1) { // If player O wins the game
        cout << "player O won!" << endl;
        scoreofO++; // Increments the score of O
    }
    else if (check_tie()) { // if game is tied
        cout << "It's a draw!\n\n";
    }
}

void ComputerChoice(int ComputerRow, int ComputerCol) { // function to determine the
computers choice of input.
pcin: // statements will carry out until it reaches goto
    if (winforO() == 0) { //checks if there is any winning move available for the
computer
        srand(time(NULL)); // to generate random number between 1 & 3
        ComputerRow = rand() % 3 + 1; // stores random number as the row number for
computer
        ComputerCol = rand() % 3 + 1; // stores random number as the column number
for computer
        if (board[ComputerRow - 1][ComputerCol - 1] != '?') { // checks if the
position selected is empty
            goto pcin; // jumps back to beginning of the statements if position
is not empty
        }
        else if (board[ComputerRow - 1][ComputerCol - 1] == '?') { // if position
is empty
            board[ComputerRow - 1][ComputerCol - 1] = 'O'; // inserts O at that
position
            cout << "the computer chose (" << (ComputerRow) << "," <<
(ComputerCol) << ")" << endl;
        } // Prints the row and column number chosen by the user
    }
}

void playerInput(int PlayerRow, int PlayerCol) { // Function to determine the players
choice of row and col input
in1: // statements used as a loop to enter correct row number
    if (winforX() == 0) { //checks if there is any winning move available for the
player
        cout << "Enter a row (1, 2 or 3) for player X: "; // asks user to input a
row num

```

```

        cin >> PlayerRow;
        if (PlayerRow > 3 || PlayerRow < 1) { // validates if number input is
within range 1 & 3
            cout << "wrong input, try again" << endl; // Asks user to input row
again in case of wrong input
            goto in1; // jumps back to beginning
        }
        else if (PlayerRow <= 3 && PlayerRow >= 1) { // if row is correct it
proceeds to column number input
            in2: // new set of statements used as a loop to enter correct col number
                cout << "Enter a column (1, 2 or 3) for player X: "; // asks user to
input a col num
                cin >> PlayerCol;
                if (PlayerCol > 3 || PlayerCol < 1) { // validates if number input
is within range 1 & 3
                    cout << "wrong input, try again" << endl; // Asks user to
input col again in case of wrong input
                    goto in2; // jumps back to beginning
                }
                else if (PlayerCol <= 3 && PlayerCol >= 1) { // if column num also
correct
                    if (board[PlayerRow - 1][PlayerCol - 1] == '?') { // checks
wheteher position is empty
                        board[PlayerRow - 1][PlayerCol - 1] = 'X'; // inserts
symbol X in that position if place is empty
                    }
                    else {
                        cout << "this cell is already taken, try again" <<
endl; // else outputs this message
                        goto in1; // jumps back to beginning of the fuction
                    }
                }
            }
        }
    }

void startGame(bool player[3][3], bool computer[3][3]) { // Human vs Computer game)
    int PlayerRow = 0, PlayerCol = 0, ComputerRow = 0, ComputerCol = 0;
gamestart:
    Display_Board(); // displays the board
    playerInput(PlayerRow, PlayerCol); // Calls the function for human player to
insert row and col for next move
    if (winforX() == 1) { // Checks if there is a winning possibility for Player
        printWinOrDraw(); // Prints the output based on the result of the previous
function
        Display_Board(); // Displays the board of the winning game
        cout << "X now has a score of " << scoreofX << endl; // Outputs the score
of Player
    }
    else if (winforO() == 1) { // Checks if there is a winning possibility for
Computer
        printWinOrDraw(); // Prints the output based on the result of the previous
function
        Display_Board(); // Displays the board of the winning game
        cout << "O now has a score of " << scoreofO << endl; // Outputs the score
of Computer
    }
}

```



```

    else {
        ComputerChoice(ComputerRow, ComputerCol); // computer enters its choice
        cout << endl;
        goto gamestart; // jumps back to start of the function
    }
}

int main() // main interface of the game
{
    int x_score = 0, o_score = 0, score = 1, choice, round = 0;
    int ComputerRow, ComputerCol;
    int PlayerRow, PlayerCol;
    cout << "Welcome to Tic Tac Toe game. Do you want to play against: " << endl;
    cout << "1. Friend" << setw(15) << "2. Computer" << endl;
    cin >> choice;
    if (choice == 1) { // If player chose to play with another player
        while (x_score != 3 && o_score != 3) // loop to check if either of the
        players got a score of 3
        {
            board_reset(); // resets the board for every new game

            while (!HumanvsHuman_check_win() && !check_tie()) // continues the
            game until either of the players won or the game is tied
            {
                Display_Board(); // Displays the board
                HumanvsHuman_check(); // Checks all possibilities for a player
                to win
            }

            if (check_tie()) // checks if the game is tied
            {
                cout << "The game is tie" << endl;
            }
            else if (mark == 'O') // If HumanvsHuman_check returns 'O', then it
            means Player X won
            {
                cout << 'X' << " wins" << endl;
                x_score++; // increments the score of Player X
            }
            else if (mark == 'X') // If HumanvsHuman_check returns 'X', then it
            means Player O won
            {
                cout << 'O' << " wins" << endl;
                o_score++; // increments the score of Player O
            }
            round++; // Increments the number of rounds of games played

            cout << "Game " << round << " is over" << endl;
            cout << 'X' << " score is " << x_score << endl;
            cout << 'O' << " score is " << o_score << endl;
            cout << "*****" << endl;
        }
        cout << "thanks for playing" << endl; // Outputs the final details on the
        screen
    }
    else if (choice == 2) { // If the user chose to play with the computer

```

```

        while (scoreofX != 3 && scoreofO != 3) { // Loop to check if either the
computer or player won 3 games
            startGame(player, computer); // Carry out the game
            board_reset(); // reset the board before every new game starts.
        }
        if (scoreofO == 3) { // if Player 'O's score is 3 , computer wins
            cout << "the computer won the game" << endl;
        }
        else if (scoreofX == 3) { // if Player 'X's score is 3 , player wins
            cout << "you won the game" << endl;
        }
        cout << "Thanks for playing!" << endl; // games ends
    }
    system("pause>0");
}

```

### **Brief Description about the functions:**

- **Display\_Board():-**

This function will render a Board layout for TIC TAC TOE.

- **HumanvsHuman\_check():-**

This function checks whose turn it is. If it is 'X' then it will print the message for 'X' otherwise for 'O'. And ask the user to choose the row and column where the 'X' or 'O' will get printed. Also, it checks if the user chooses a valid place.

- **HumanvsHuman\_check\_win():-**

This function checks if a player won or not. First it will check the columns. Then it will check the rows. And finally, It will check diagonally. And if a player has won then it will return true otherwise it will return False.

- **check\_tie():-**

This function checks whether both players met the tie or not. This simple function checks every row and column for the '?' sign. If it is found then it will return true otherwise it will return false.

- **board\_reset():-**

This function clears each and every choice made by users in the game. It clears out any 'X' or 'O' signs in the game and finally, restarts the game to get a clean board layout again.

- **ComputerChoice (int ComputerRow,int ComputerCol):-**

This function produces random numbers so that the computer performs as a player in the game. This function contains srand and rand. rand() functions to generate random numbers.

· **printWinorDraw():-**

This function checks who won the match or if there is a draw for the players. It checks whose score is higher, if there is an equal score then the match is a draw.

· **playerInput(int PlayerRow, int PlayerColumn):-**

This simple and large function is doing nothing but taking the players input for the game. It is checking if the input is valid or not. It is also checking if the player chooses an empty block or not.

· **startGame():-**

This function starts the game by calling appropriate functions like DisplayBoard() etc. First it will take the input and put it in the board and print the board, and simultaneously it is checking if someone won or not or if it is a tie.

· **int main():-**

The most important function is here. It is declaring the important variables and printing the directional messages for the players. Player 1 is the user and the other is the Computer or it can be player 2 instead. And in the switch, it is resetting if there is a tie, otherwise it prints the input in the board and after all “if else” it will print the final result.

· **int winfor0():-**

This function is used to check whether “0” won the game or not. They check it along the diagonal, horizontal and vertical. Returns “true” if “0” won, or else returns false. Used in Human Vs Computer part of the program.

· **int winforX():-**

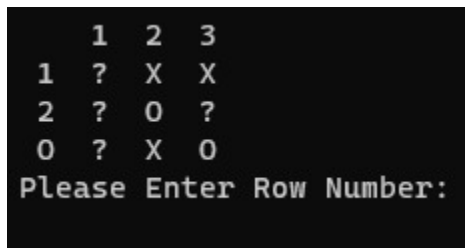
This function is used to check whether “X” won the game or not. They check it along the diagonal, horizontal and vertical. Returns “true” if “X” won, or else returns false. Used in Human Vs Computer part of the program.

## **Errors, Difficulties & Solutions:**

### ***Errors:***

1. When we called array elements, we had forgotten to decrement by 1 the row and the column number. Hence there were many logical errors. Since array values should always be subtracted by 1.
2. After compiling the entire program, when we were checking at the very end, we found that in computer vs human subdivision of the program, after round 2 the computer never gave any inputs. And the game never concluded, hence the game continued without an end.

3. The display board showed “0” instead of “3” in the row numbers, even though initially it had shown “3”, and after a few rounds it displayed “0”.



```
  1  2  3
1  ?  X  X
2  ?  O  ?
0  ?  X  O
Please Enter Row Number:
```

4. The code for HumanVsComputer had considered “?” as a filled position. Hence, we were not able to input any “X” or “Y” characters into it.
5. “0” was never able to win in the human vs human subdivision of the program. After three inputs of “X”, the game was declared to be a tie.

### ***Difficulties:***

The “Human vs Computer” was done by a pair coordinating their work together, and the Human vs Human was done by another set of pair coordinating their work together too. Since each pair used their own logic, it was hard for the compiler to compile programs with completely different logic.

Creating the code for finding the best of three rounds took a lot of our time.

### ***Solutions:***

‘O’ was unnecessarily placed in the wrong position in the display of tic tac toe due to the error in the array declared. Hence when the computer chose a random position, the “O” (“O” being computer’s character) was placed along the numberings of rows and columns.

The question mark was considered as a filled position by the program due to the human error we made in the while loop. Then we had to change the while loop condition, the condition then stated that as long as the row and column number was within the range 1-3, the character will be inscribed in the code.

We created two new functions: winforO() and winforX(). These functions let the user and computer continue input characters until one of them wins, or until there is a draw. In this way we were able to overcome the problem that “0” can never win.

Many errors were caused when we used the initially used the board array declared globally for the computer vs human program. Hence, we created two new arrays for the subdivision of the program, human vs computer. The arrays were computer[3][3] and player[3][3].

## **Distribution of Work:**

- **Yasmine:**

Worked on the design and layout of the report, provided output screenshots, as well as worked on the code and comments of the human vs computer part of the program.

- **Sara:**

Worked on the human vs human subdivision of the program. Nullified the errors present in both the human vs human and human vs computer subdivisions of the program. Created the layout of the report.

- **Jannat:**

Compiled both the subdivisions; “human vs human” and “human vs computer” into one program. Worked on human vs computer subdivision of the program. Created the initial layout of the output of the program. Nullified the errors present in both the subdivisions of the program.

- **Rabaa:**

Worked on human vs computer subdivision of the program. Worked on giving various screenshots of the output and creating the lab report.

- **Hana:**

Worked on the human vs computer subdivision of the program. Worked on giving the description of each function, and the introduction of the lab report. The brief description of the errors, difficulties faced and solutions to the errors of the program were stated.

- **Teamwork:**

We sat together as a team to understand the method we should use to create the program. The websites and YouTube videos we took help to create this program were chosen together. And finally placed all our work into one lab report. And prepared for the presentation together.

## **Output Screenshots:**

### ***Human VS Human:***

*When Player ‘O’ score is 3:*

```

Welcome to Tic Tac Toe game. Do you want to play against:
1. Friend    2. Computer
1
  1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column: 3
1
  1 2 3
1 ? ? ?
2 ? ? ?
3 X ? ?
Choose where to insert your 'O' mark, insert row then column: 2
2
  1 2 3
1 ? ? ?
2 ? O ?
3 X ? ?
Choose where to insert your 'X' mark, insert row then column: 3
3
  1 2 3
1 ? ? ?
2 ? O ?
3 X ? X
Choose where to insert your 'O' mark, insert row then column: 1
1
  1 2 3
1 O ? ?
2 ? O ?

```

```

Choose where to insert your 'X' mark, insert row then column: 3
1
  1 2 3
1 X X O
2 ? O ?
3 X ? ?
Choose where to insert your 'O' mark, insert row then column: 2
1
  1 2 3
1 X X O
2 O O ?
3 X ? ?
Choose where to insert your 'X' mark, insert row then column: 3
3
  1 2 3
1 X X O
2 O O ?
3 X ? X
Choose where to insert your 'O' mark, insert row then column: 2
3
0 wins
Game 1 is over
X score is 0
O score is 1
*****
  1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column:

```

```

Choose where to insert your 'X' mark, insert row then column: 1
3
  1 2 3
1 ? ? X
2 ? 0 ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column: 1
2
  1 2 3
1 ? 0 X
2 ? 0 ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column: 1
1
  1 2 3
1 X 0 X
2 ? 0 ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column: 3
2
0 wins
Game 3 is over
X score is 0
O score is 2
*****
  1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column:

```

```

1 X ? ?
2 ? ? ?
3 0 ? ?
Choose where to insert your 'X' mark, insert row then column: 1
2
  1 2 3
1 X X ?
2 ? ? ?
3 0 ? ?
Choose where to insert your 'O' mark, insert row then column: 1
3
  1 2 3
1 X X 0
2 ? ? ?
3 0 ? ?
Choose where to insert your 'X' mark, insert row then column: 2
1
  1 2 3
1 X X 0
2 X ? ?
3 0 ? ?
Choose where to insert your 'O' mark, insert row then column: 2
2
0 wins
Game 4 is over
X score is 0
O score is 3
*****
thanks for playing

```

*When Player 'X' enters selected position:*

```

 1 2 3
1 ? ? ?
2 ? X ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column: 1
3
 1 2 3
1 ? ? 0
2 ? X ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column: 3
1
 1 2 3
1 ? ? 0
2 ? X ?
3 X ? ?
Choose where to insert your 'O' mark, insert row then column: 3
3
 1 2 3
1 ? ? 0
2 ? X ?
3 X ? 0
Choose where to insert your 'X' mark, insert row then column: 3
3
Already selected, try another position

```

*When there is a tie:*

```

Choose where to insert your 'O' mark, insert row then column: 1
1
 1 2 3
1 0 ? 0
2 ? X X
3 X ? 0
Choose where to insert your 'X' mark, insert row then column: 1
2
 1 2 3
1 0 X 0
2 ? X X
3 X ? 0
Choose where to insert your 'O' mark, insert row then column: 2
1
 1 2 3
1 0 X 0
2 0 X X
3 X ? 0
Choose where to insert your 'X' mark, insert row then column: 3
2
The game is tie
Game 2 is over
X score is 0
O score is 1
*****
 1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column:

```

*When invalid row or column is entered:*



```

 1 2 3
1 ? 0 X
2 ? 0 ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column: 1
1
 1 2 3
1 X 0 X
2 ? 0 ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column: 3
2
O wins
Game 3 is over
X score is 0
O score is 2
*****
 1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column: 1
1
 1 2 3
1 X ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column: 3
3
Choose a valid row and column between 1 and 3, try again:

```

*When Player 'X' wins the game:*

```

3
 1 2 3
1 0 ? X
2 ? 0 ?
3 X ? X
Choose where to insert your 'O' mark, insert row then column: 2
3
 1 2 3
1 0 ? X
2 ? 0 0
3 X ? X
Choose where to insert your 'X' mark, insert row then column: 3
3
Already selected, try another position
 1 2 3
1 0 ? X
2 ? 0 0
3 X ? X
Choose where to insert your 'X' mark, insert row then column: 3
2
X wins
Game 1 is over
X score is 1
O score is 0
*****
 1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column:

```

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Choose where to insert your 'O' mark, insert row then column: 1
3
  1 2 3
1 0 ? 0
2 ? X ?
3 ? ? ?
Choose where to insert your 'X' mark, insert row then column: 3
2
  1 2 3
1 0 ? 0
2 ? X ?
3 ? X ?
Choose where to insert your 'O' mark, insert row then column: 3
1
  1 2 3
1 0 ? 0
2 ? X ?
3 0 X ?
Choose where to insert your 'X' mark, insert row then column: 1
2
X wins
Game 2 is over
X score is 2
O score is 0
*****
  1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Choose where to insert your 'O' mark, insert row then column:
X score is 1

```

### ***Computer VS Human:***

*When Player 'X' wins the game (scores 3 points):*

Welcome to Tic Tac Toe game. Do you want to play against:

1. Friend    2. Computer

2

1 2 3

1 ? ? ?

2 ? ? ?

3 ? ? ?

Enter a row (1, 2 or 3) for player X: 1

Enter a column (1, 2 or 3) for player X: 1

the computer chose (2,3)

1 2 3

1 X ? ?

2 ? ? 0

3 ? ? ?

Enter a row (1, 2 or 3) for player X: 3

Enter a column (1, 2 or 3) for player X: 1

the computer chose (1,2)

1 2 3

1 X 0 ?

2 ? ? 0

3 X ? ?

Enter a row (1, 2 or 3) for player X: 2

Enter a column (1, 2 or 3) for player X: 1

player X won!

1 2 3

1 X 0 ?

2 X ? 0

3 X ? ?

Enter a row (1, 2 or 3) for player X: 2

Enter a column (1, 2 or 3) for player X: 1

player X won!

1 2 3

1 X 0 ?

2 X ? 0

3 X ? ?

X now has a score of 1

1 2 3

1 ? ? ?

2 ? ? ?

3 ? ? ?

Enter a row (1, 2 or 3) for player X:

```

3 ? 0 ?
Enter a row (1, 2 or 3) for player X: 3
Enter a column (1, 2 or 3) for player X: 1
the computer chose (2,3)

  1 2 3
1 ? ? ?
2 ? X 0
3 X 0 ?
Enter a row (1, 2 or 3) for player X: 1
Enter a column (1, 2 or 3) for player X: 2
the computer chose (1,1)

  1 2 3
1 0 X ?
2 ? X 0
3 X 0 ?
Enter a row (1, 2 or 3) for player X: 1
Enter a column (1, 2 or 3) for player X: 3
player X won!
  1 2 3
1 0 X X
2 ? X 0
3 X 0 ?
X now has a score of 2

```

```

1 ? ? ?
2 ? ? ?
3 ? ? ?
Enter a row (1, 2 or 3) for player X: 1
Enter a column (1, 2 or 3) for player X: 3
the computer chose (3,3)

  1 2 3
1 ? ? X
2 ? ? ?
3 ? ? 0
Enter a row (1, 2 or 3) for player X: 2
Enter a column (1, 2 or 3) for player X: 2
the computer chose (3,2)

  1 2 3
1 ? ? X
2 ? X ?
3 ? 0 0
Enter a row (1, 2 or 3) for player X: 3
Enter a column (1, 2 or 3) for player X: 1
player X won!
  1 2 3
1 ? ? X
2 ? X ?
3 X 0 0
X now has a score of 3
you won the game
Thanks for playing!

```

*When invalid row or column is entered:*

```
2 ? ? ?
3 ? ? ?
Enter a row (1, 2 or 3) for player X: 1
Enter a column (1, 2 or 3) for player X: 3
the computer chose (3,2)

  1 2 3
1 0 X X
2 ? ? ?
3 ? 0 ?
Enter a row (1, 2 or 3) for player X: 2
Enter a column (1, 2 or 3) for player X: 2
the computer chose (2,1)

  1 2 3
1 0 X X
2 0 X ?
3 ? 0 ?
Enter a row (1, 2 or 3) for player X: 2
Enter a column (1, 2 or 3) for player X: 3
the computer chose (3,3)

  1 2 3
1 0 X X
2 0 X X
3 ? 0 0
Enter a row (1, 2 or 3) for player X: 3
Enter a column (1, 2 or 3) for player X: 5
wrong input, try again
```

*When the computer scores some points:*

```
3 ? ? ?
Enter a row (1, 2 or 3) for player X: 1
Enter a column (1, 2 or 3) for player X: 2
the computer chose (2,1)

  1 2 3
1 0 X X
2 0 0 X
3 ? ? ?
Enter a row (1, 2 or 3) for player X: 3
Enter a column (1, 2 or 3) for player X: 1
the computer chose (3,3)

  1 2 3
1 0 X X
2 0 0 X
3 X ? 0
Enter a row (1, 2 or 3) for player X: 3
Enter a column (1, 2 or 3) for player X: 2
player O won!
  1 2 3
1 0 X X
2 0 0 X
3 X X 0
O now has a score of 1
```

*When there is a tie:*

There will never be a tie, either X wins or O wins.

```

3 X ? ?
Enter a row (1, 2 or 3) for player X: 2
Enter a column (1, 2 or 3) for player X: 1
the computer chose (1,1)

  1 2 3
1 0 ? 0
2 X ? ?
3 X ? ?
Enter a row (1, 2 or 3) for player X: 2
Enter a column (1, 2 or 3) for player X: 2
the computer chose (1,2)

  1 2 3
1 0 0 0
2 X X ?
3 X ? ?
Enter a row (1, 2 or 3) for player X: 3
Enter a column (1, 2 or 3) for player X: 2
player 0 won!
  1 2 3
1 0 0 0
2 X X ?
3 X X ?
0 now has a score of 2
  1 2 3

```

*When user selects taken position:*

```

Welcome to Tic Tac Toe game. Do you want to play against:
1. Friend    2. Computer
2
  1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Enter a row (1, 2 or 3) for player X: 2
Enter a column (1, 2 or 3) for player X: 1
the computer chose (3,2)

  1 2 3
1 ? ? ?
2 X ? ?
3 ? 0 ?
Enter a row (1, 2 or 3) for player X: 3
Enter a column (1, 2 or 3) for player X: 2
this cell is already taken, try again

```

***Additional:***

*When user doesn't choose any one of the two game modes:*

```
Welcome to Tic Tac Toe game. Do you want to play against:
1. Friend    2. Computer
3
please select either option 1 or 2 :)
Welcome to Tic Tac Toe game. Do you want to play against:
1. Friend    2. Computer
2
  1 2 3
1 ? ? ?
2 ? ? ?
3 ? ? ?
Enter a row (1, 2 or 3) for player X: 2
Enter a column (1, 2 or 3) for player X: 1
the computer chose (3,2)
```



